

04. Glomerular Cell Biology and Podocytopathies · assets 1-5 / 5

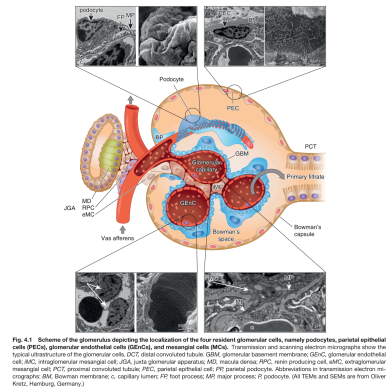
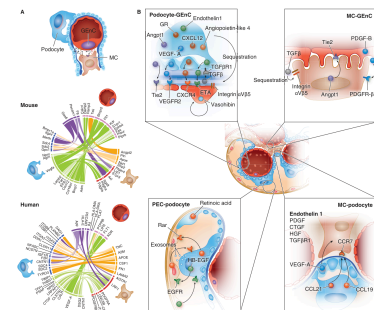
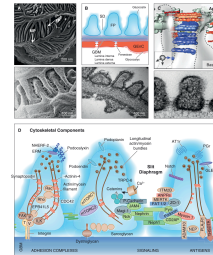


Fig. 4.1 Scheme of the glomerulus depicting the localization of the four resident glomerular cells, namely podocytes, parietal epithelial cells (PECs), glomerular endothelial cells (GEC), and mesangial cells (MC). Transmission and scanning electron micrographs show the typical ultrastructure of the glomerular cells. DGC, distal convoluted tubule; GSM, glomerular basement membrane; GEC, glomerular endothelial cell; MC, intraglomerular mesangial cell; JGA, juxta glomerular apparatus; MD, macula densa; APC, renin producing cell; eMC, extraglomerular mesangial cell; PCT, proximal convoluted tubule; PEC, parietal epithelial cell; PT, parietal podocyte. Abbreviations in transmission electron micrographs: BM, Bowman membrane; c, capillary lumen; FF, foot process; MP, major process; P, podocyte. (All TEMs and SEMs are from Oliver Kneitz, Hamburg, Germany.)

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Fig_4_4

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Fig_4_2

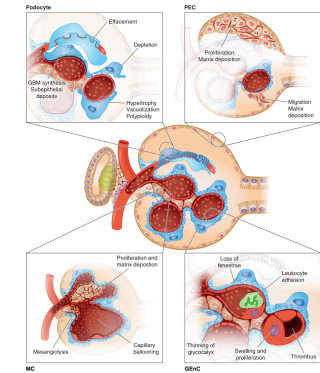


Fig. 4.3 Pathophysiological reaction patterns of glomerular cells. GEnC, glomerular endothelial cell; MC, mesangial cell; PEC, parietal epithelial cell.

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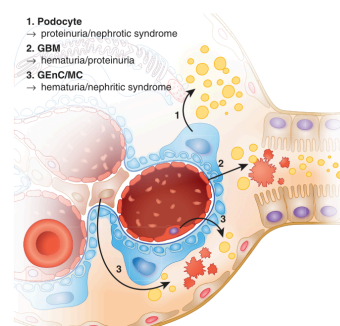


Fig. 4.5 The clinical presentation reflects the localization of a glomerular injury. Injury of podocytes results in proteinuria (yellow droplets) eventually leading to the clinical manifestation of nephrotic syndrome. Injury of the glomerular basement membrane (GBM) commonly results in proteinuria and hematuria with dysmorphic erythrocytes. Injury of glomerular endothelial cells (GenC) and mesangial cells (MC) usually leads to hematuria with dysmorphic erythrocytes and little proteinuria.

Fig_4_5